Al-baraa Hegaz

AI Engineer & Data Scientist

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Summary

Analytical fourth-year Physics and Mathematics undergraduate specializing in Artificial Intelligence, Machine Learning, and Data Science. Proficient in Python, EDA, Data Analytics, Machine Learning, Deep Learning, and AI Engineering, with experience in Excel, Power BI, and Tableau for data analysis and visualization. Seeking opportunities in Data Science, AI/ML Engineering, or Data Analytics.

Education

Al-Azhar University **Sep 2023 – Jun 2027**
Bachelor of Science in Physics and Mathematics *Cairo, Egypt*

- Relevant Coursework: Advanced Linear Algebra, Advanced Calculus, Differential Equations, Statistical Physics, and Machine Learning.

Experience

Creativa Innovation Hub **Jul 2026 – Jul 2026**
AI Hackathon *Menofia, Egypt*
AI Innovators Team | Top 5 Finalist

- Built a Hybrid RAG system combining vector search and BM25 retrieval for clinical information support.
- Developed an Agentic RAG workflow with LangGraph for query analysis, retrieval, and grounded generation.
- Implemented safety guardrails and evaluated system performance using RAGAS and LangSmith.

Neuromatch **Jul 2026**
Deep Learning Trainee *Remote*

- Completed 68-hour training in neural networks, CNNs, Transformers, generative AI, and reinforcement learning.
- Completed intensive training in deep learning theory and practical implementation using Python and PyTorch.
- Studied modern deep learning techniques including regularization, attention mechanisms, unsupervised learning, and self-supervised learning.

National Telecommunication Institute (NTI) **Feb 2026 – Apr 2026**
Data Analysis Trainee *Menofia, Egypt*

- Completed hands-on training in SQL Server, Microsoft Excel, DAX, Power BI, Tableau, and Python.
- Built interactive dashboards and visualizations in Power BI and Tableau to communicate data-driven insights.
- Performed data cleaning, analysis, and reporting to support data-driven decision-making.

2B Academy **Jul 2025 – Sep 2025**
Data Scientist Track *Cairo, Egypt*

- Completed the Data Scientist Track within the Digital Transformation Department.
- Applied data science techniques to develop analytical solutions and generate data-driven insights.

Projects

Sadeed | AI-Powered Clinical RAG System

- Built a Hybrid RAG system combining vector search and BM25 retrieval for clinical information support.
- Developed an Agentic RAG workflow using LangGraph for query analysis, retrieval, and grounded generation.
- Implemented safety guardrails and evaluated system performance using RAGAS and LangSmith.

Fahman.ai | Socratic Arabic Reading Tutor for Kids

- Built a Socratic AI tutor that diagnoses Arabic reading errors and guides children aged 6–9 to decode words.
- Integrated Nile-Chat-4B, Qari-OCR, and Whisper with FastAPI and Next.js, with a rule-based question-bank fallback.
- Developed the business plan and ranked 16th out of 139 teams (Top 12%) at ImpactX 2026.

Food Mart Sales Analysis | Interactive Power BI Dashboard

- Analyzed Food Mart sales data using Power BI, performing data cleaning and exploratory data analysis (EDA).
- Designed an interactive dashboard with KPIs for revenue, cost, profit, margins, customer count, and returns.
- Developed regional, product, and time-based analyses to identify performance trends and business opportunities.

Skills

Programming: Python, SQL, Object-Oriented Programming (OOP)
AI & Machine Learning: Machine Learning, Deep Learning, Scikit-learn, PyTorch, Natural Language Processing (NLP), Transformers, RAG, AI Engineering
Data Science & Analytics: Data Analysis, Exploratory Data Analysis (EDA), Pandas, NumPy, Statistics, Feature Engineering
Data Visualization: Power BI, Tableau, Microsoft Excel, Matplotlib, Seaborn
Tools & Platforms: Git, GitHub, Jupyter Notebook, FastAPI, LangGraph, LangSmith, RAGAS

Certifications

Associate AI Engineer for Developers (Career Track) — DataCamp | Certificate — Jun 2026

Supervised Machine Learning: Regression and Classification — Coursera | Certificate — Aug 2026

Python for Data Science and Machine Learning Bootcamp — Udemy | Certificate — Nov 2025

McKinsey.org Forward Program — McKinsey & Company | Certificate — Jun 2026

InnovEgypt Entrepreneurship Program — ITIDA & TIEC | Certificate — Aug 2026

Leadership & Community

Organizer, 10th International Conference of the Faculty of Science — Al-Azhar University **2026**
Conference Organizer *Cairo, Egypt*

- Contributed to organizing an international scientific conference connecting students, researchers, and academics.
- Engaged with research presentations spanning Artificial Intelligence, Machine Learning, Physics, and scientific innovation.

Volunteer, College Student Activities — Al-Azhar University, Faculty of Science **2025**
Volunteer *Cairo, Egypt*

- Supported student welcoming events and contributed to organizing university activities.

Executive Board Member, National Al-Azhar Student Union **2021 – 2022**
Student Representative *Egypt*

- Contributed to strategic planning and collaborated with student teams across multiple governorates.
- As Menofia Governorate president, represented hundreds of students and organized events, building leadership skills.
- Developed leadership, public speaking, communication, and teamwork skills.

Languages

Arabic: Native **English:** Professional Working Proficiency **German:** Intermediate